Week 11 — Multi-Omics Paper Analysis Assignment

Instructions: Choose any THREE of the four papers. Answer all three questions for each paper, citing relevant figures, analytical methods, and key results.

Paper 2 — A Universal Gut-Microbiome-Derived Signature Predicts Cirrhosis

Cell Metabolism, 2020 | Metagenomics + Metabolomics + Machine Learning

1. Analysis Workflow: How were shotgun metagenomic and metabolomic data processed to identify disease-associated features?

Answer. The study is a two-tier nested design rather than a single case-control comparison, and this structure drives every downstream choice. The proband tier contains 27 patients with NAFLD-cirrhosis (16 compensated, 11 decompensated) and 54 non-NAFLD controls, while the second tier contains 82 first-degree relatives of both groups, of whom 51 had no NAFLD, 21 had NAFLD without advanced fibrosis, 3 had NAFLD-related fibrosis and 7 had cirrhosis. Phenotypes were assigned by MRI proton density fat fraction of at least 5% for NAFLD, by biopsy stage 4 or magnetic resonance elastography of at least 4.69 kPa for cirrhosis, and by elastography of at least 3.63 kPa for advanced fibrosis; transient elastography above 11.8 kPa was substituted in the 11 relatives in whom MRI was contraindicated. The two tiers answer different questions. The probands provide a severe, hospital-based training set, whereas the relatives share household, diet and genetic background but are mostly at an earlier disease stage, so the relative cohort tests both sensitivity to early disease and whether the signature is confounded by family structure. Stool and fasting serum were collected in parallel and stored at −80 °C.

Metagenomic processing. DNA was extracted with the Earth Microbiome Project protocol (MagAttract PowerSoil), libraries were prepared with Nextera XT in quarter reactions, normalized to 8 pM with 1% PhiX and sequenced on a MiSeq. Reads were trimmed with skewer, human reads were removed with Bowtie2 against hg19, taxonomy was assigned with MetaPhlAn2 v2.7.7, and function was profiled with HUMAnN2 v0.11.2, which maps reads to ChocoPhlAn pangenomes and translates the remainder with DIAMOND against UniRef90 before quantifying MetaCyc pathways; tables were renormalized with humann2_renorm_table and imported through curatedMetagenomicData. The resulting feature space was large: 1,216,335 gene families, 7,355 pathways, 310 species across 53 families and 115 genera, and 188 strains. This asymmetry, thousands of features against 81 training samples, is the central statistical constraint of the study and explains why feature selection had to be aggressive.

Why the analysis is ordered as it is. Community-level structure was tested before any feature-level test, using inverse Simpson alpha diversity and weighted-UniFrac PCoA with PERMANOVA ($P < 0.001$; Fig. 1b,c). This ordering matters, because it establishes that the groups differ at the level of the whole community and therefore that feature-level differences are

not isolated artefacts of multiple testing. Differential abundance was then computed with DESeq2 rather than with LEfSe or MaAsLin. The choice is defensible: DESeq2 models counts with a negative binomial distribution, shares dispersion across features and returns calibrated adjusted p-values, whereas the LDA score used by LEfSe is an effect-size heuristic without a null distribution. The strongest single effect was an 8.6-fold increase of Veillonella dispar (adjusted $P = 4.79 \times 10^{-43}$). Of 310 species, 108 correlated with clinical metadata by Spearman analysis (Extended Data Fig. 2), and 39 of 7,355 pathways differed, concentrated in aromatic amino acid biosynthesis, branched-chain amino acid biosynthesis, nucleotide biosynthesis and cell wall metabolism (Fig. 2A). Cross-cohort pathway concordance was quantified with RV coefficients rather than by pooling samples, and fibrosis-stage comparisons used Kruskal-Wallis tests with a zero-inflated negative binomial mixed model, which is appropriate for sparse, over-dispersed taxa.

Metabolomics. Four extraction solvents were benchmarked by base-peak chromatogram complexity and 50% methanol was selected; 50 µL of extract were combined with 150 µL of methanol containing an ampicillin internal standard, separated on a 30 x 2.1 mm C18 column over an 840 s gradient at 0.5 mL per min, and analysed on a Bruker MaXis qTOF with top-10 CID fragmentation and dynamic exclusion. Processing in MS-DIAL 3.66 used 0.01 Da MS1 tolerance, 0.025 Da MS2 tolerance and 0.2 min retention-time alignment, followed by manual inspection to remove false positives, and annotation in MS-FINDER 3.20 required 10 ppm mass error and an MS2 score of at least 8 out of 10, although metabolites of interest were accepted at 7.5 to 8.0 after manual review. This yielded 435 metabolites, of which 75 correlated with clinical metadata. It is worth stating plainly that 159 features remained unidentified, so the metabolite signature is chemically informative only for the annotated minority.

Harmonizing external cohorts. The validation cohorts were not taken from published summary tables. Italian raw fastq files were reprocessed with dada2 and RDP taxonomy, and the Chinese and mixed-fibrosis cohorts were re-derived through curatedMetagenomicData, so that every cohort passes through a comparable pipeline. This is the correct approach when platform and pipeline differences cannot be modelled as a batch effect, but it has a consequence the authors acknowledge: the Italian cohort is 16S-based and therefore genus-resolved, so the species-level signature had to be applied at genus level there.

Key evidence: Fig. 1A–D; Fig. 2A,B; Fig. 3A; Extended Data Fig. 2; STAR Methods (cohort, metagenomics, metabolomics, statistical analysis).

2. Predictive Modeling: How were microbial and metabolite signatures selected, modeled, and validated using random forests and ROC/AUC?

Answer. Feature selection was performed twice, independently, on the two molecular layers, and this is the design decision that gives the paper its central claim: a microbial signature and a metabolite signature were derived without reference to each other, and each reached an AUC of 0.91.

Feature selection. Within each layer two filters were intersected. DESeq2 identified features that differed between cirrhosis and control, and a random forest ranked the same features by Mean Decrease Gini importance; the intersection gave 19 species (Fig. 1E) and, separately, 17 metabolites (Fig. 3A). Importance ranking is not decorative here. With compositional microbiome data, many taxa co-vary because they share ecology, so a set chosen purely on significance tends to include redundant features that inflate apparent performance in small samples; ranking by importance and taking the intersection removes that redundancy. The 19 species are biologically coherent, combining enrichment of oral-origin Veillonella parvula, V. atypica, V. dispar, Acidaminococcus sp. D21 and Ruminococcus gnavus with depletion of butyrate producers Eubacterium rectale, E. eligens, Faecalibacterium prausnitzii and Dorea longicatena.

Model implementation and validation. Random forests were fitted with the train function of the R caret package under 10-fold cross-validation, and ROC curves and AUCs were computed with pROC. The validation strategy is the strongest element of the design: instead of a random train and test split, the model was trained only on the 27 probands and 54 controls and then applied unchanged to four external sets, namely the 82 first-degree relatives, a Chinese cohort (123 cirrhosis versus 114 controls, total 237), an Italian cohort, and a mixed-aetiology fibrosis cohort. This tests transfer across ethnicity, diet, sequencing batch and disease prevalence, and it is considerably more demanding than internal cross-validation. Performance held: AUC 0.91 (95% CI 0.89–0.93) in training, 0.88 (0.72–1.00) in relatives, 0.86 (0.82–0.91) in the Chinese cohort and 0.89 (0.80–0.98) in the Italian cohort (Fig. 1F; Fig. 5A,B).

What the clinical increments mean. Age was added because it is the dominant risk factor and is available everywhere, and adding it improved discrimination in the relative cohort. Albumin was added because it is a marker of hepatic synthetic function and, critically, was the only clinical variable recorded in every cohort, so it is the only one that can be part of a transferable model; with 22 features the AUC rose to 0.91 in relatives and 0.95 in the Chinese cohort (Fig. 5C,D). AST was reserved for the clinically harder question of separating cirrhosis from NAFL or from mild-to-moderate fibrosis in the mixed cohort, where the species model alone gave 0.85 and 0.84 and AST raised these to 0.94 and 0.91 (Fig. 5E–H). The pattern is instructive: microbial features carry information that clinical chemistry does not, and clinical chemistry carries information that is complementary rather than redundant.

Why two signatures are better than one ensemble. The metabolite model was deliberately not merged into the final classifier. Merging would have produced a single number but no independent confirmation; keeping the layers separate means that two entirely different molecules measured on different instruments converge on the same diagnosis, which is a much stronger form of internal replication. The drop from 0.91 in training to 0.86 on external data also quantifies the optimism of the training estimate, and it is honest to report it rather than to quote the training AUC alone.

Limitations. Four limitations should be stated. The training set is small (81 samples for 19 features), no FDR threshold is reported for the DESeq2 step, and the random-forest hyperparameters and tree counts are not given, so exact reproduction is not possible. The cohort is single-centre and the compensated and decompensated groups are pooled despite different prognosis. Finally, the Italian cohort size differs between the text and the figure legend (46 versus 35 cirrhotic patients), and the relative cohort contains only seven cirrhosis cases, which is why its confidence interval spans 0.72 to 1.00; the point estimate alone should not be over-interpreted.

Key evidence: Fig. 1E,F; Fig. 3A,B; Fig. 5A–H; Extended Data Fig. 6D; STAR Methods (random forest, ROC analysis).

3. Multi-Omics Integration: How were microbial pathways, metabolites, and clinical parameters integrated to identify cirrhosis-associated signatures?

Answer. Integration proceeded through three layers, pathway, then metabolite, then clinical, on the principle that each layer must independently point at the same biology before a diagnostic signature is proposed.

Pathway layer. The 39 discriminant MetaCyc pathways were not treated as an unordered list but grouped into coherent modules: aromatic amino acid biosynthesis (chorismate and 3-dehydroquinate routes), branched-chain amino acid biosynthesis (L-valine), nucleotide biosynthesis (UMP, IMP and adenosine ribonucleotides), cell wall synthesis, and starch and glycolysis (Fig. 2A). Mapping pathways back to the genera that carry them identified Acidaminococcus and Veillonella as the organisms responsible for the amino acid fermentation signal, which is consistent both with their known physiology and with the species-level result, and which turns a statistical list into a metabolic story: a shift from saccharolytic, butyrate-producing fermentation towards amino acid fermentation with ammonia and hydrogen production. Concordance of these pathway profiles with the Chinese and mixed-fibrosis cohorts was quantified by RV coefficients, which is the appropriate statistic when samples cannot be pooled.

Metabolite layer. The 17 discriminators read as the chemical shadow of the same shift: aromatic amino acid derivatives (tryptophan, indole, indole-3-carboxaldehyde, indole-3-carboxylic acid), branched-chain amino acid dipeptides (leucyl-proline, γ-glutamyl-isoleucine), the primary bile acid glycochenodeoxycholic acid, the sphingolipid C18-sphingosine, and reduced kynurenic acid and vitamin D3 species (Fig. 3A,F). The tryptophan result is worth spelling out because it is mechanistically informative rather than merely differential: microbial dissimilation products such as indole increased while the host kynurenine pathway product kynurenic acid decreased, so total tryptophan rose because it was being shunted towards microbial metabolism. That is a two-compartment explanation of a single metabolite ratio, and it is exactly the kind of inference that requires both omics layers to be measured in the same samples.

Network construction and experimental triangulation. The layers were joined by a Spearman correlation network of 19 species against 17 metabolites, then extended with the gene families of the discriminant species into a 33-node, 338-edge network (Fig. 4A,B). This is deliberately a hypothesis-generating structure, and one edge was tested: Ruminococcus gnavus strain VPI C7–9 was cultured anaerobically across a 12-point chenodeoxycholic acid gradient (0.001 to 5000 μg/mL), lipids were extracted by the Folch method, and both C18-sphingosine and 1H-indole-3-carboxaldehyde increased with bile acid concentration (Fig. 4C). A second edge, between Faecalibacterium prausnitzii and kynurenic acid, rests on correlation alone. The asymmetry is honest: the network proposes many relationships and only one was experimentally supported.

Clinical integration as sensitivity analysis. Clinical parameters entered last and only as an add-one-at-a-time analysis in which BMI, albumin, ALT, AST, bilirubin, cholesterol, GGT, glucose, insulin and platelet count were each added to the 19-species base model (Extended Data Fig. 6D). This is a defensible way to handle clinical data, because it asks which measurement adds independent information instead of allowing clinical variables to drive the classifier. It also produces the paper's most practical conclusion, namely that the signature is 19 species plus age, optionally plus albumin, with AST reserved for the cirrhosis-versus-fibrosis comparison.

Critical appraisal. Three points limit the interpretation. First, there is a partial circularity in adding albumin, since hypoalbuminaemia is itself a consequence of cirrhosis, so part of the gain reflects disease severity rather than new diagnostic information. Second, 159 of the metabolic features remained unidentified, so the metabolite branch of the signature is not fully chemically defined and would be difficult to translate into a clinical assay. Third, and most fundamental, the design is cross-sectional and observational: the network and the bile acid experiment establish plausibility, not that the microbiome causes cirrhosis, a point the authors state explicitly.

Key evidence: Fig. 2A,B; Fig. 3A,F; Fig. 4A–C; Extended Data Fig. 2, 6D.

Paper 3 — Shared and Disease-Specific Host Gene–Microbiome Associations

Nature Microbiology, 2022 | Transcriptomics + Microbiomics

1. Analysis Workflow: How were paired host RNA-seq and 16S rRNA sequencing data processed and compared across CRC, IBD, and IBS?

Answer. The defining design choice is that the microbiome was sampled from colonic mucosal biopsies rather than stool and paired with host RNA-seq from the same tissue, so the association under test is between tissue-adherent microbes and the epithelium they contact. This is a materially different question from stool-based studies, because the mucosal community is closer to the host interface and is not dominated by luminal transit effects.

Cohorts and pairing. The dataset comprises 208 pairs, 416 samples, drawn from three published studies. Colorectal cancer contributes 88 pairs from 44 individuals, each providing tumour and matched adjacent normal tissue, so the comparison is within-person and controls for

host genotype and environment, although the adjacent tissue is itself a field-effect tissue rather than a true healthy control. Inflammatory bowel disease contributes 78 pairs, comprising 56 IBD (34 Crohn's disease, 22 ulcerative colitis) and 22 non-IBD controls from the HMP2 cohort. Irritable bowel syndrome contributes 42 pairs, 29 patients and 13 healthy controls, with some individuals sampled at two time points and averaged. The imbalance in disease prevalence across cohorts (7 of 82 in the relative tier of the CRC study, for example, versus 29 of 42 in IBS) is important when comparing statistical power between diseases.

Why cohorts were processed separately. Host transcriptomes were processed within each cohort and the pipelines differ substantially: the CRC data were newly generated with stranded total RNA on a HiSeq 2500, Trimmomatic trimming, HISAT2 against hg38 and featureCounts (mean alignment 87.11%), while IBD used a Broad Picard pipeline and IBS used TopHat-based MAP-RSeq against hg19. The 16S data differ as well, with V5-V6 regions clustered through QIIME in CRC, V4 through UPARSE in IBD and V4 aligned with BURST in IBS. The authors therefore did not merge the cohorts or apply batch correction; they analysed each cohort end to end and compared results afterwards. This is the correct decision, because standard batch-correction methods assume a shared feature space and a common technical model, neither of which holds here. The cost is that cross-disease statements are comparisons of results rather than of samples. As a robustness check, DADA2 amplicon variants were generated for a subset of CRC samples and compared with the OTU profiles, giving concordance of 0.89 at phylum level and 0.60 at genus level, so the main pipeline is validated but not replaced.

Feature matrices. Within each cohort, genes were restricted to protein-coding transcripts, filtered to those expressed in at least 50% of samples, variance-stabilized with DESeq2 and filtered at the 25th percentile of variance, retaining 12,513 genes in CRC, 11,985 in IBD and 12,429 in IBS. On the microbial side, archaea, chloroplast and reagent contaminants were removed, taxa were aggregated at six taxonomic levels and filtered for a relative abundance of 0.001 in at least 10% of samples. The six levels were then concatenated and centre-log-ratio transformed. Concatenation means that reads from the same clade are counted several times and that features are strongly collinear, which the authors acknowledge and mitigate by using penalized methods with stability selection; they also compared concatenate-then-CLR with CLR-per-level-then-concatenate and showed agreement (mean Pearson r = 0.92; Extended Data Fig. 5), which is the evidence that makes this non-standard transformation acceptable. The result was 235 taxa in CRC, 121 in IBD and 238 in IBS, the low IBD count reflecting the well-documented diversity loss in that disease.

Global correspondence test. Before searching for individual associations, the authors asked whether the two layers correspond globally, using Procrustes analysis on Aitchison distances for host genes and Bray-Curtis distances for microbes with 9,999 permutations, corroborated by Mantel tests. Only CRC showed correspondence ($P = 1 \times 10^{-4}$; Mantel P = 0.0026), whereas IBD (P = 0.1; Mantel P = 0.26) and IBS (P = 0.42; Mantel P = 0.95) did not (Fig. 1b). This negative result is the justification for sparse, subset-to-subset methods: if the whole transcriptome does

not track the whole microbiome, one must search for subspaces that do. It also carries a caution that is often missed, since the absence of global correspondence does not establish the absence of local associations, only that they do not dominate the overall structure.

Sampling caveats. Two caveats belong to the workflow itself. The 88 CRC pairs derive from 44 individuals, so the pairs are not independent observations and the effective sample size is smaller than the nominal one, a form of pseudoreplication that inflates apparent power for that disease. And because only 16S composition was measured, with no shotgun functional profiling, every functional statement in the paper is an inference from taxonomy rather than a measurement of microbial gene content.

Key evidence: Fig. 1a,b; Extended Data Fig. 4,5; Methods (host RNA-seq, preprocessing of gene expression and microbiome data).

2. Integration Algorithms: How were sparse CCA, LASSO regression, and pathway enrichment used to identify host–microbe associations?

Answer. Three algorithms were used in a deliberate hierarchy, from modules to individual gene-microbe pairs, because the datasets differ in dimension by roughly two orders of magnitude (about 12,000 genes against 100 to 240 taxa) and both are highly collinear. The hierarchy is the point: sparse CCA answers which gene modules co-vary with which microbial modules, and LASSO answers which individual gene is associated with which individual taxon once covariates are controlled.

Sparse CCA for modules. Sparse CCA was implemented with the PMA package, which places L1 penalties on both canonical vectors so that only a few genes and few taxa load on each component, which is what makes the components interpretable under p much greater than n. Penalties were tuned by grid search with leave-one-out cross-validation rather than by PMA's built-in permutation search, for a concrete technical reason: that search optimizes only one penalty at a time, whereas the grid covers both dimensions. The selected values were 0.15 for taxa and 0.20 for genes in CRC, 0.177 and 0.333 in IBD, and 0.40 and 0.10 in IBS. Ten components were computed per cohort, separately for cases and controls, and component significance was assessed on held-out cross-validated scores by correlation test with Benjamini-Hochberg FDR below 0.1, giving 7 significant components in CRC, 4 in IBD and 6 in IBS (Fig. 2). The unit of inference is the component rather than the feature, which is why the authors can speak of a host pathway associating with a set of microbes. The IBS solution is noticeably less sparse than the others, with 61 taxa per component on average against 6 to 8 in the other two diseases, which is worth noting because it weakens the claim of a focused microbial module in IBS.

LASSO for gene-level inference. LASSO models were fitted gene by gene, with each host gene as the response and all taxa plus sex and disease subtype (Crohn's versus ulcerative colitis; IBS-C versus IBS-D) as predictors, using glmnet with leave-one-out cross-validation for lambda.

Because ordinary LASSO coefficients are shrunk and cannot yield valid p-values, inference used the desparsified lasso from the hdi package to obtain confidence intervals and p-values, followed by FDR correction. Stability selection with stabs was then applied, using 100 subsamples of half the data with the penalty perturbed around its optimum and a selection frequency threshold of 0.6. An association was retained only if it passed both stability selection and FDR below 0.1, yielding 755 associations in CRC, 1,295 in IBD and 441 in IBS (Fig. 3a). The computational scale is worth stating because it explains the design: roughly 12,300 gene models against about 200 taxa per cohort, taking on average five and a half hours on five nodes.

Why cases and controls were modelled separately. The authors did not add a disease interaction term, and they give the reason: interaction terms would double the predictor count to roughly 400 per model and, more importantly, would impose a shared sparsity structure on two groups that may not share one. They instead fitted separate models and then demonstrated that case and control associations show zero overlap even without any significance threshold, and again after down-sampling to equal sample sizes. This is what licenses the term disease-specific throughout the paper, and it is a cleaner argument than a significance test on a difference.

Enrichment and its specificity rule. Pathway enrichment used Fisher exact tests with Benjamini-Hochberg FDR below 0.1, on MSigDB canonical pathways: KEGG plus PID for the sparse CCA results, and KEGG plus PID plus Reactome for the LASSO results, with size and overlap filters of 25 to 300 genes and at least 5 overlapping for CCA, and 10 to 85 genes and at least 5 for LASSO. In practice only about 99 pathways per cohort survived these filters, so the enrichment is best described as targeted rather than genome-wide, and the authors acknowledge that informative pathways were certainly discarded. Disease specificity of a pathway required either significance in cases with no significance in controls, or, where both were significant, a significant difference in enrichment at FDR below 0.2. That second clause is necessary, because otherwise every shared immune pathway would be mislabelled as specific.

Critical appraisal. Four methodological limitations should be weighed. There is no GSEA, so the analysis is set-based over-representation rather than a ranking-based test, and no GO terms are used. Most of the 1,881 candidate pathways were discarded by the size filters. An FDR of 0.1 is lenient by convention, although it is applied on top of stability selection at a 0.6 frequency threshold, which is the more conservative of the two filters. And because only taxonomic composition was measured, the enrichment results describe host biology associated with taxa, not with microbial functions.

Key evidence: Fig. 1a; Fig. 2a–d; Fig. 3a–e; Methods (sparse CCA, lasso regression, stability selection, pathway enrichment).

3. Biological Interpretation: How were shared and disease-specific regulatory networks identified, and what biological conclusions can be drawn?

Answer. The structure that emerges is specific and somewhat counter-intuitive: the microbial taxa associated with host genes are largely shared between diseases, whereas the host pathways they associate with are largely disease-specific. The same organisms appear in different diseases, coupled to different host biology.

Shared pathways with disease-specific partners. Only three host pathways were significantly microbe-associated in all three diseases, of which the two named in the text are oxidative phosphorylation and the RAC1 pathway (Fig. 2a), representing mitochondrial energy metabolism and a cytoskeletal and immune signalling module involved in epithelial restitution. Against these three sit 102 disease-specific pathways, distributed as 52 in CRC, 25 in IBD and 25 in IBS. The illustrative case is RAC1 (Fig. 2b): the same host gene set, including CYBB, NCF1, CYFIP2, STAT3, CTNNB1 and PAK2, is associated with entirely different organisms in each disease, namely oral-origin Streptococcus, Synergistales and GN02 in CRC, Granulicatella and Clostridium sensu stricto 1 in IBD, and Bacteroides massiliensis, Bifidobacterium and Odoribacter in IBS. The host module is conserved and the microbial partner is not, which argues that the functional output of a microbe is context-dependent rather than an intrinsic property of the taxon.

Disease-specific networks. At pathway level, CRC is distinguished by the Syndecan-1 pathway associating with Parvimonas and Bacteroides fragilis (Fig. 2c), IBD by integrin β1 signalling associating with Peptostreptococcaceae, Intestinibacter and Phascolarctobacterium, which is coherent with leukocyte recruitment into inflamed mucosa, and IBS by ribosomal genes. At gene level, IL-10 pathway genes such as CXCL8 and IL1RN correlate negatively with Anaerolineae and TM7 in CRC, MAPK3 and VIPR1 correlate with Peptostreptococcaceae in IBD, and in IBS Prevotella correlates negatively with SMC5 while B. massiliensis correlates positively with PLA2G4A, linking a taxon to arachidonic acid metabolism. The LASSO level yields 18 further disease-specific pathways, distributed 4 in CRC, 9 in IBD and 5 in IBS, including interleukin-10 signalling and NOTCH1 signalling in CRC, thrombin signalling through PARs in IBD, and SUMOylation of DNA damage response proteins and PPAR signalling in IBS. These are mechanistically distinct and mutually exclusive, which is what a meaningful disease-specific association should be.

Shared nodes and their rewiring. The shared components were quantified: three microbes associate with host genes in all three diseases, namely Peptostreptococcaceae, Streptococcus and Staphylococcus, with 20 microbes shared between at least two diseases, while five host genes are shared across all three and 135 between at least two (Fig. 4a,b). PINK1 is the clearest case of rewiring, associating with Collinsella in CRC, Peptostreptococcaceae in IBD and Blautia in IBS, three organisms with very different ecology, and PINK1 itself sits at the intersection of mitochondrial homeostasis and PI3K/AKT signalling, which is plausible for all three diseases for different reasons. ANXA1 and TLR4 behave similarly, and the CRC-IBD shared gene set is enriched for natural killer cell mediated cytotoxicity and leucocyte transendothelial migration.

Biological model. The resulting model has two tiers. There is a common susceptibility skeleton of immunoregulatory and metabolic pathways, including oxidative phosphorylation, RAC1, VEGF, complement and coagulation, and cytokine-receptor interaction, that the microbiome engages in all three diseases; but in each disease it is engaged by a different set of organisms, so the downstream output differs. This is consistent with the clinical picture, in which IBD is dominated by IL-23/IL-17 signalling, CRC by Notch and WNT activation, and IBS by altered antibacterial gene expression and purine salvage, and it offers a mechanism for why dysbiosis looks similar across diseases while the host consequences do not.

Critical appraisal. Four limits constrain how far this can be pushed. The data are cross-sectional, so direction cannot be resolved, and disease-altered mucosa may simply permit colonisation by particular taxa rather than being altered by them. The apparent strength of microbial specificity is partly a technical artefact, since 84% of host genes are shared as input features across diseases whereas only 37% of taxa are, and disease-specific genes are present as inputs elsewhere 80% of the time against 12% for taxa. Different 16S regions and pipelines limit cross-disease comparability of taxa, and no functional metagenomics was performed, so all mechanisms remain inferred from taxonomy and from prior literature rather than demonstrated here.

Key evidence: Fig. 2a–d; Fig. 3a–e; Fig. 4a,b; Extended Data Fig. 3; Supplementary Tables 19–22.

Paper 4 — α-Ketoglutarate Signaling to Chromatin

Science, 2026 | Functional Genomics + Metabolomics + Epigenomics + Transcriptomics

1. Experimental Workflow: How was the αKG biosensor developed and integrated with CRISPR screening to identify metabolic regulators?

Answer. The problem is one of measurement rather than of genetics: α-ketoglutarate is a small, rapidly turning-over metabolite that serves as a co-substrate for nuclear dioxygenases, and no method existed to read out the nuclear pool specifically in living human cells. Existing FRET-based tools had been used largely in bacteria, and classical fractionation cannot resolve a pool that is consumed within minutes. The solution is a transcriptional biosensor, and the engineering logic of that sensor determines the specificity of the screen that follows.

Sensor architecture. The sensing element is NtcA, a cyanobacterial transcription factor that responds to nitrogen status by binding α-ketoglutarate, which increases its affinity for DNA (structure PDB 8H40). NtcA was converted into a chimeric activator consisting of a VP64 activation domain, NtcA and two SV40 nuclear localization signals under an EF1alpha promoter, and paired with a reporter in which five tandem NtcA binding sites derived from the glnA and glnN promoters drive minimal-CMV-controlled GFP. Optimization was empirical and iterative. Nuclear and cytoplasmic fractionation with lamin B1 and GAPDH controls showed that early

constructs were partly cytoplasmic and partly truncated; a NetNES1.1 prediction identified a cryptic nuclear export signal inside VP64; removing the VP16 monomer improved nuclear localization but reduced output; and replacing the activation domain and swapping the myc NLS for two SV40 NLSs produced the final sensor (Fig. 1A,D; Extended Data Fig. 1). This is a useful illustration of a general point, namely that a metabolite sensor must be validated for localization before its readout can be interpreted as a compartment-specific measurement.

Performance envelope. Performance was characterized quantitatively. Microscale thermophoresis gave a Kd of 2.0 mM for α-ketoglutarate binding to NtcA, and the sensor resolves roughly 0.5 to 6 mM whole-cell α-ketoglutarate; in HEK293 cells, 0.3 to 3 mM dimethyl-α-ketoglutarate shifted the GFP-positive fraction from about 10% to about 70% (Fig. 1C,E). Specificity was controlled with two dead variants, an effector-binding mutant (R90E) and a DNA-binding-domain deletion, and by showing no response to (R)-2-hydroxyglutarate or dimethyl succinate and no reversal of the IDH1-R132H effect by the mutant-IDH inhibitor vorasidenib, which excludes a 2-hydroxyglutarate artefact (Extended Data Fig. 4). Two confounders were also mapped rather than ignored: both sensor arms increased in G2/M, presumably through reporter copy number, and dying cells showed autofluorescence. Both are precisely the kind of artefact that a dual-arm screen is designed to cancel.

Dual-arm CRISPR screen. A custom CRISPR-Cas9 knockout library targeting 127 genes that directly synthesize, catabolize or transport α-ketoglutarate was introduced into U251 glioblastoma cells carrying the reporter, at a multiplicity of infection of 0.3 with 3×10^6 cells per arm and three replicates. The decisive feature is the dual-arm format: a functional sensor and an inactive R90E sensor were screened in parallel, so any sgRNA that changes GFP for reasons unrelated to α-ketoglutarate is subtracted rather than selected. On day 10, cells in the top and bottom 10% of GFP were sorted, sgRNA cassettes sequenced, and genes ranked with MAGeCK and Apron (Fig. 3A). Four genes emerged: BCAT1 and DLD, whose loss increased nuclear α-ketoglutarate, and GPT2 and SLC25A11, whose loss decreased it (Fig. 3B,C). Each was validated with two independent sgRNAs, immunoblotting for protein depletion and sgRNA-resistant cDNA rescue (Fig. 3D–K; Extended Data Fig. 6).

Critical appraisal. The main limitation is coverage: 127 genes is a targeted library, chosen because a genome-wide screen would be dominated by essential genes and by generic transcriptional effects, but it means the screen can only find genes already annotated as acting on α-ketoglutarate, and it cannot discover indirect regulators. The authors also report a case in which this mattered, since OGDH scored poorly because its chronic loss is toxic, even though short-term depletion still raised nuclear α-ketoglutarate. Screening was performed in a single cell line, U251, with confirmatory work in normal human astrocytes and TS516 cells, and FACS-based selection remains sensitive to growth-rate differences even with the inactive arm as a control.

Key evidence: Fig. 1A–E; Fig. 2; Fig. 3A–C; Extended Data Fig. 1, 4, 6, 7; Methods (biosensor engineering, CRISPR screen).

2. Multi-Omics Analysis: How were metabolomic, ChIP-seq, and transcriptomic results analyzed to connect αKG metabolism with chromatin regulation?

Answer. The integration here is unusually strict: the epigenomic layer defines the gene set, the transcriptomic layer tests whether that gene set matters, and pharmacological rescue supplies directionality. The layers are not merely correlated, they are used in sequence to narrow and then to test a hypothesis.

Metabolomics as ground truth. Targeted LC-MS used a ZIC-pHILIC column with 10 mM ammonium acetate at pH 9.8 against an acetonitrile gradient on a Q Exactive HF-X or Exploris 480 at 120,000 to 240,000 resolution, with metabolites extracted in ice-cold 80% acetonitrile and normalized to total ion current. Absolute quantification used isotope dilution against $^{13}C_4$-α-ketoglutarate at a 200 nM internal standard with a ten-point calibration curve and 1/X squared weighting, and cell number was measured so that concentrations could be expressed per cell volume, giving 1.35 mM whole-cell α-ketoglutarate in U251 cells. ^{15}N-glutamine tracing with the alanine M+1 to glutamate M+1 ratio measured GPT2 flux directly (Fig. 4F), and DNA methylation was quantified by UHPLC-MS/MS with transitions for 5-methyldeoxycytidine and 5-hydroxymethyldeoxycytidine, supplemented by single-base-resolution enzymatic sequencing (Fig. 4C; Fig. 5D). The point of the absolute measurement deserves emphasis: without a concentration, the finding could not be compared with the Km of the demethylases, and that comparison turns out to be central to the interpretation.

ChIP-seq with spike-in normalization. ChIP-seq was performed on brain tissue at postnatal day 19 for H3K4me3, H3K9me3 and H3K27me3 with Drosophila chromatin and a spike-in antibody for normalization. Spike-in calibration is not optional in this experiment: the phenotype is a global increase in methylation, which internal normalization would partly cancel and which input normalization could mistake for a loading difference. Libraries were sequenced to 30, 50 and 40 million reads respectively; reads were aligned with Bowtie2 to mm10, peaks called with MACS2 using broad settings for H3K9me3 and H3K27me3 and narrow settings for H3K4me3 at FDR below 0.1, differential peaks identified with DiffBind, and distribution shifts tested by Kolmogorov-Smirnov tests (Fig. 6A–C).

Integration by intersection. The integration step is the intersection in Fig. 6D. Differential H3K27me3 peaks were intersected with down-regulated genes, giving 884 genes that are simultaneously hypermethylated and repressed, and differential H3K4me3 peaks were intersected with up-regulated genes, giving more than 1,200 genes (Fig. 6D; Extended Data Fig. 15A). This is a much stronger filter than either layer alone, because it requires the chromatin change and the expression change to occur at the same locus in the same direction, and it converts a genome-wide methylation phenotype into a defined gene set. Enrichment of that set gave nervous system development, neurogenesis, gliogenesis and regulation of calcium concentration (Fig. 6E), and GSEA showed coordinated loss of the brain H3K27me3-marked gene set (NES = -2.07, FDR 0.001) and of cranial nerve development genes (NES = -2.18), with

individual loci such as Hapln4, Etv4 and Notch3 confirmed by genome browser tracks (Fig. 6J–O).

Ascorbate and the methylation readout. Two details separate a real epigenetic mechanism from a generic chromatin change. First, the hypermethylation phenotype required ascorbate supplementation, the cofactor for TET and JmjC demethylases, and without it the effect was not observed (Fig. 4B); this ties the phenotype specifically to dioxygenase chemistry, since a structural chromatin change would not be ascorbate-dependent. Second, the global 5-methylcytosine to cytosine ratio increased but did not reach significance in bulk measurement ($P = 0.23$), whereas single-base-resolution sequencing detected a significant increase. That discrepancy is not a contradiction but a sensitivity difference, since a change distributed across many CpGs is diluted in a bulk ratio; reporting only the bulk result would have produced a false negative.

Rescue as the directionality test. Administering dimethyl-α-ketoglutarate to Gpt2 knockout mice shifted the transcriptome back towards wild type, reducing differentially expressed genes from 494 up and 343 down to 69 up and 116 down (Fig. 5I–K), and normalized the specific loci above. The combination of quantification, tracing, chromatin profiling and rescue means each link in the chain is measured rather than assumed. The remaining limitations are technical: peak counts and fold changes are presented graphically rather than as explicit numbers, the RNA-seq analysis pipeline is attributed to a commercial provider rather than specified, and ChIP-seq was performed on bulk brain, so cell-type composition is a possible confounder that the study does not formally exclude.

Key evidence: Fig. 4B,C,F; Fig. 5B–K; Fig. 6A–O; Extended Data Fig. 15; Methods (metabolomics, ChIP-seq, RNA-seq analysis).

3. Mechanistic Validation: How did genetic perturbation, epigenomic profiling, and αKG rescue experiments establish the GPT2–SLC25A11 regulatory mechanism?

Answer. The mechanism was established by epistasis, by compartment-resolved isoforms and by rescue at three scales, chromatin marks, transcription and organismal phenotype. Each of these addresses a different weakness in the preceding evidence.

Epistasis. Overexpressing GPT2 raised nuclear α-ketoglutarate only when SLC25A11 was present, and GPT2 overexpression could not rescue the hypermethylation caused by SLC25A11 loss, which places SLC25A11 downstream of GPT2 (Fig. 4D; Extended Data Fig. 12). This is the single experiment that converts a co-occurrence of two screen hits into a pathway, and it is worth noting that the logic is asymmetric: the failure to rescue, not the success, is the informative result.

Isoform resolution. The authors then resolved an ambiguity that would otherwise have undermined the model. U251 cells and normal human astrocytes predominantly express the transcript ENST00000340124.9, which carries an N-terminal mitochondrial targeting sequence,

together with a minor isoform that lacks it and would be nucleocytoplasmic. Only the mitochondrial isoform raised nuclear α-ketoglutarate when overexpressed, indicating that the relevant synthesis happens inside mitochondria and that the nuclear GPT2 pool described previously is not the source (Extended Data Fig. 11). Without this experiment, the whole subcellular logic of the paper would rest on the assumption that GPT2 is mitochondrial.

A parallel branch through DLD. DLD was shown to act in parallel rather than in series. DLD loss still raised nuclear α-ketoglutarate in GPT2-null cells and rescued their methylation phenotype, consistent with DLD acting through the oxoglutarate dehydrogenase complex rather than through GPT2 (Fig. 4E). The tracing data agree: DLD loss reduces GPT2 flux towards pyruvate, while GPT2 loss abolishes alanine labelling entirely (Fig. 4F). The negative controls are equally informative, since loss of PDHA1 or BCAT2 produced no such rescue, which excludes a general effect on pyruvate or branched-chain amino acid metabolism.

Chromatin and mouse phenotypes. In cells, loss of either GPT2 or SLC25A11 increased H3K4, H3K9 and H3K27 methylation, and the effect depended on ascorbate (Fig. 4B). In mice, a systemic Gpt2 knockout generated by gene trap (KOMP project CSD24977, C57BL/6N background, backcrossed to C57BL/6J and then to CD-1) showed broad hypermethylation of all three marks in brain at postnatal day 19, with other organs largely spared apart from heart H3K9 and spleen H3K9 and H3K27 (Fig. 5A; Extended Data Fig. 13). The organ specificity is itself an argument, because it matches the neurological phenotype of GPT2 deficiency and because the brain is where the developmental gene programme is most dependent on these marks.

Rescue at three scales. At the cellular level, re-expression or cDNA rescue normalized histone methylation. At the pharmacological level, dimethyl-α-ketoglutarate given intraperitoneally at 600 mg per kg per day from postnatal day 0 normalized all three marks in knockout brain (Fig. 5E–H), restored the body-weight deficit (Fig. 5L) and reverted the transcriptional signature towards wild type. At the tumour level, isogenic HCT116 xenografts differing only in GPT2 status were treated with dimethyl-α-ketoglutarate, which normalized methylation without changing tumour growth rate, excluding a general fitness artefact (Extended Data Fig. 14). Using growth-matched controls for the tumour experiment is the detail that makes it convincing, because a methylation change in a slower-growing tumour would have been uninterpretable.

Boundary condition and final model. One boundary condition defines the model: dimethyl-α-ketoglutarate did not demethylate chromatin in GPT2-wild-type cells. Nuclear α-ketoglutarate is therefore not rate-limiting for demethylases under normal supply and becomes limiting only when the GPT2-SLC25A11 route is interrupted. This resolves an apparent paradox, since the Km of these enzymes for α-ketoglutarate is 5 to 60 μM, well below the measured millimolar concentration; the authors attribute the discrepancy to competitive inhibition by structurally related metabolites. The resulting mechanism is a supply chain: mitochondrial GPT2 transaminates glutamate with pyruvate to produce α-ketoglutarate and alanine; SLC25A11 exports it to the cytosol as part of the malate-aspartate shuttle; cytosolic α-ketoglutarate either enters the nucleus for TET and JmjC enzymes or is consumed by BCAT1. Breaking the chain at

either step lowers nuclear α-ketoglutarate, silences neurodevelopmental genes through hypermethylation, and produces a phenotype that an α-ketoglutarate ester can reverse, connecting an inborn error of metabolism to a chromatin-level therapeutic hypothesis.

Key evidence: Fig. 4A–F; Fig. 5A–L; Fig. 6; Extended Data Fig. 11–14.

References

Oh, T.G., Kim, S.M. et al. A universal gut-microbiome-derived signature predicts cirrhosis. Cell Metabolism 32, 1–11 (2020).

Priya, S. et al. Identification of shared and disease-specific host gene-microbiome associations across human diseases. Nature Microbiology 7, 780–795 (2022).

Sternisha, A.C., Li, H. et al. A transcriptional biosensor reveals mechanisms of α-ketoglutarate signaling to chromatin. Science 393, eadx8675 (2026).